

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

System F: Voice-Based Virtual Assistant for Multilingual Customer Support

Task 1: PEAS Specification

The selected system is a Voice-Based Virtual Assistant (VVA) that handles customer support queries across multiple languages in real time web chat, and mobile platforms.

Performance Measures

1. Customer query resolution rate
2. Speech recognition accuracy
3. Customer satisfaction score (CSAT)
4. Average response time
5. Language translation accuracy
6. First-contact resolution rate

Environment

- Customer service call centers
- Mobile applications and websites
- Noisy and quiet environments
- Users speaking different languages and accents
- Internet and telecommunication networks

Actuators

The agent modifies its environment through the following output mechanisms:

- Text-to-Speech (TTS) engine
- Automated voice response system
- Chat response generator
- Ticket creation and routing system
- Notification and email sender

Sensors

The agent perceives its environment through the following input sources:

- Microphone input
- Speech recognition system
- Language detection module
- Customer profile database
- Conversation history logs

Task 2: Environment Classification

Dimension	Choice	Justification
Observability	Partially Observable	The assistant cannot fully know the customer's intentions, emotions, or background knowledge.
Agents	Multi-agent	The system interacts with customers and sometimes human support agents.
Determinism	Stochastic	Similar customer inputs can produce different outcomes due to speech variations and uncertainty.
Episodic vs Sequential	Sequential	Earlier conversation context affects future responses and decisions.
Static vs Dynamic	Dynamic	Customer requirements and conversation states change continuously during interaction.
Discrete vs Continuous	Continuous	Voice signals, speech patterns, and timing are continuous in nature.

Task 3: Critical Analysis

1. Biggest Technical Problem

The biggest problem is that we can't always understand what customers mean. They might not speak clearly use words that're hard to understand switch between languages or not give us all the information we need. We have to try to figure out what they want from the bit of information they give us which can be tricky. This makes it hard to give them answers especially when they are speaking different languages.

2. Utility Function

Utility Function:

$$U = 0.7 \times \text{Accuracy} + 0.3 \times \text{ResponseSpeed}$$

Where:

- Accuracy = correctness of the response
- ResponseSpeed = speed of providing the response

If the weight of Accuracy is doubled:

$$U = 1.4 \times \text{Accuracy} + 0.3 \times \text{ResponseSpeed}$$
$$U = 1.4 \times \text{Accuracy} + 0.3 \times \text{ResponseSpeed}$$

The assistant will make sure to give accurate answers to customer requests. This means the assistant will take a little time to think about the answers. The service will be better because of this. However it may take the assistant a bit longer to get to the customers. The assistant thinks that giving answers is more important than giving fast answers. So the assistant will take the time it needs to give the possible answers to customer requests.

Task 4: Structured Data Design — Structured Representation
separate file in Json